

1 Strategic Planning of Agroforestry Supply Chains

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8 Abstract

9 Agroforestry systems integrate fast-growing woody perennials into agricultural
10 land. They simultaneously deliver multiple biomass fractions of differing quality to
11 food, chemical, pulp-and-paper, as well as energy industries. The woody biomass
12 feedstock produced can be used to substitute fossil feedstocks in chemical and
13 energy industry. Additionally, agroforestry increases climate resiliency and sus-
14 tainability of agricultural production. In contrast to woody biomass from forestry,
15 agroforestry-based woody biomass shows substantial feedstock potentials as most
16 land is used for agriculture in central Europe. Despite substantial environmental
17 and economic potentials, agroforestry is only applied in marginal scale in central
18 Europe. A core challenge in fostering agroforestry is to establish economically
19 viable regional bioeconomy networks which connect partners from agriculture as
20 well as downstream industries. Such an agroforestry-based supply chain needs to
21 be structured and organized in a way that allows all partners to benefit economi-
22 cally in the long run. This paper presents a supply chain design model that helps
23 to support design decisions for agroforestry-based supply chains. To this end, it
24 models agroforestry establishment and harvest scheduling on farm-level in a linear
25 multi-echelon network design model. It relies on biomass growth models which
26 describe the production of multiple woody biomass types in agroforestry sys-
27 tems. These biomass fractions are pretreated and transported to consumers from
28 chemical, pulp-and-paper, and energy industry depending on quality and demand
29 restrictions. The model is applied in a case study focusing on central Germany
30 where a heterogeneous industrial cluster is located within a fertile agricultural

production area. The optimization results indicate that agroforestry-based supply chains can be configured in an economically viable way. Agroforestry systems can serve as feedstock suppliers for industrial consumers in the same region at a considerable scale. Profitability of agroforestry-based agricultural practice seems to be competitive with traditional agricultural production and at the same time allow producing woody biomass at a competitive cost, provided transports are not too long.

Keywords: Industrial symbiosis, Agroforestry, Biomass cascading, Bioeconomy network design, Supply chain optimization, Biobased production

Introduction

Industrial bio-economy depends on connecting primary biological production systems to multiple industrial sectors through networks that efficiently allocate biomass fractions of varying quality to their highest-value application (Hildebrandt et al., 2019). Agroforestry systems (AFS) that deliberately integrate fast-growing woody perennials into agricultural land represent realizations of this principle. By co-producing woody biomass fractions (like stems, branches, twigs, leaves etc.), AFS create diverse feedstocks that can serve multiple industrial consumer sectors like chemical, pulp-and-paper industry, as well as energy industry Trujillo et al. (2026). This biological-industrial symbiosis substitutes fossil feedstocks with renewable bio-based alternatives and stores biogenic carbon in bio-based chemical products Montagnini & Nair (2004).

Despite their ecological and industrial potentials, temperate AFS occupy only a marginal share of agricultural land in practice. In Germany, approximately 1,200 hectares of silvoarable AFS were established in 2025, accounting for barely 0.02 % of total agricultural land Deutscher Fachverband für Agroforstwirtschaft (DeFAF) e.V. (2026). This gap between supposed potentials and observed practices indicates implementation challenges which often root in economic, organizational, and regulatory obstacles. Establishing AFS at regional scale requires coordinated decisions on stand

establishment, harvest scheduling, and biomass allocation across spatially distributed industrial consumers Low et al. (2023). For example, farmers will only invest in AFS when they are convinced that they will generate revenues from selling wood to industrial customers at the time of first harvest which is, however, 5-15 years after establishing an AFS. This long time span imposes a severe investment risk as investment costs are immediate but revenue is uncertain and far away (Millard, 2011). On the other hand, industrial consumers particularly in process industries (like chemical and pulp-and-paper industry) prefer reliable and high-volume biomass supply contracts which imposes a challenge for farmers as AFS growth depends on weather as do harvest decisions (Harayama et al., 2020). Next to harvest scheduling, many other business decisions need to be made. Early harvests promise earlier revenues, but harvested biomass is dominated by small-scale branches, twigs, and residues with limited usability in downstream industries and lower value (Kauter et al., 2003). At longer rotations, biomass quantity is higher and mostly composed of stems whose diameters surpass quality thresholds such that the share of chemical- and paper-grade material rises markedly (Huber et al., 2018; Niemczyk & Thomas, 2021). The biomass feedstock quantity and quality in an AFS-based supply chain therefore depends on the rotation strategies and the industrial demand structures of multiple farms and industrial consumers. Current biomass supply chain models, reviewed in [section 2](#), do not consider this aspect as biomass availability is usually given at exogenous time series or optimize a single downstream supply chain rather (Zahraee et al., 2020).

This paper closes this gap by formulating an AFS supply chain design (AFS-SCD) problem. The linear program integrates design and operation decisions of biomass production in AFS with multiple downstream industries via logistical processes. This way, interdependencies between biomass production decisions in AFS, logistical handling decisions, and biomass allocation decisions in downstream processes are modeled explicitly. Biomass production decisions are based on growth models which estimate

the quantity of biomass types in AFS. Specifically, biomass types are defined based on their suitability for downstream processing. Three different grades are distinguished: chemical/biorefinery grade, pulp and paper grade, and energy grade. A quality cascade is assumed: biorefinery-grade biomass can also be used in pulp-and-paper as well as energy applications, but the reverse does not hold. Depending on stand age, quantities of these grades are available in AFS which guide harvest decisions. Once harvests are scheduled, biomass flows to pre-treatment and storage hubs are planned where biomass grades are allocated and transshipments to consumers are prepared depending on demand quantity and the consumers' willingness-to-pay. This way, the proposed AFS-SCD configures a supply chain for a regional AFS-based industrial ecosystem. It helps to study effects of ecological, regulatory, and economic parameters on the profitability of AFS-based supply chains.

The remainder of the paper is structured as follows. Section 2 reviews the relevant literature on AFS, biomass growth modeling, and biomass supply chain design. Section 2 presents the mathematical formulation of the AFS-SCD. Section 2 describes a case study illustrating the application of the AFS-SCD to a real-world problem. Section 2 summarizes the findings of the paper and discusses directions for future research.

Literature review

Agroforestry biomass systems, growth models of woody biomass, and biomass supply chain modeling have each been studied extensively, but have not yet been combined in an integrated framework. This section reviews literature on (i) agroforestry systems and their economics, (ii) biomass growth models for AFS considering multiple biomass grades, and (iii) biomass supply chain design problems.

108 **Agroforestry systems**

109 Agroforestry deliberately integrates woody perennials with crops and/or livestock
110 on the same management unit, with the aim of enhancing productivity, profitability,
111 and/or environmental performance (Nair, 1993; Toensmeier, 2017). In temperate
112 climates, two crop-based system types can be distinguished based on the rotation
113 length at which woody biomass is harvested.

114 **Short-rotation agroforestry systems (SRAFS)** integrate narrow tree strips
115 (typically 1–4 rows of the same fast-growing species like poplar or willow) into an
116 otherwise arable field in an alley-cropping arrangement. Crop production continues in
117 the inter-row strips (Huber et al., 2018). Thus, tree strips typically cover 10–30% of
118 the total parcel area (Graves et al., 2007). Rotation lengths and total stand lifetime in
119 SRAFS are typically between 3–8 years.

120 **Long-rotation agroforestry systems (LRAFS)** use the same alley-cropping spatial
121 structure as SRAFS but with rotation cycles of 10–20 years or longer (Graves et
122 al., 2007; Nair, 1993). At this rotation length, individual trees develop substantial
123 stem diameters and the proportion of high-value timber increases markedly, enabling
124 premium material uses (veneer, sawlogs, engineered wood products). Therefore, also
125 timber-quality tree species (such as walnut, cherry, oak, or high-grade poplar clones)
126 along with fast growing species are planted (Reisner et al., 2007).

127 SRAFS and LRAFS are not clearly differentiated in practice. First, a clear cut-off
128 age is hard to define and, more importantly, the intended use of AFS biomass is not
129 always clear when establishing the AFS and can also change over time. Nonetheless,
130 when establishing AFS, the structure of AFS and its intended use after harvest affect
131 economic viability substantially.

132 The decision of a landowner to establish an AFS involves weighing substantial up-front
133 costs and long-term opportunity costs against a diversified revenue stream. On the cost

side, establishment requires site preparation, planting material, and planting labour, typically amounting to 1,500–3,500 €/ha for AFS (Faasch & Patenaude, 2012; Testa et al., 2014). Recurrent harvest and refitting costs (machine hire, chipping, transport to roadside) add approximately 200–400 €/ha per harvesting cycle depending on stand density and equipment availability (Spinelli et al., 2009; Testa et al., 2014). During a harvesting cycle, maintenance costs of AFS arise on a yearly basis. Next to expenses for establishment, maintenance and harvesting, opportunity costs arise as no revenues from crops are generated on tree strips. In regions with fertile soils and high cereal prices, yearly crop revenues may sum up to 400–600 €/ha (Faasch & Patenaude, 2012; Graves et al., 2007). Thus, opportunity costs are often the dominant cost driver in AFS profitability analyses.

On the benefit side, the primary revenue of AFS is the sale of harvested woody biomass, whose magnitude and frequency depend on tree ages at harvest, species, and site productivity. For poplar-based SRAFS in Central Europe, typical yields of 10 to 25 t of dry matter biomass per ha can be achieved for typical rotation lengths (Huber et al., 2018; Niemczyk & Thomas, 2021). Additionally, positive crop-yield effects in the inter-row strips of AFS may result. Tree rows act as windbreaks, reducing soil erosion and evapotranspiration. They provide partial shading that moderates heat stress during drought years and their root systems may improve water retention in top-level soil (Graves et al., 2007; Grünwald et al., 2007). Empirical measurements in Central European alley-cropping trials indicate that these microclimate effects can increase cereal yields in adjacent strips by 5–15% relative to open-field controls (Graves et al., 2007). Beyond direct yield effects, AFS deliver broader ecosystem services (like soil organic matter accumulation, carbon sequestration, biodiversity enhancement) that may lead to agri-environment subsidies and carbon credits which have to be considered as additional financial benefits (Faasch & Patenaude, 2012; Toensmeier, 2017).

160 Nonetheless, economic attractiveness of AFS depends primarily on revenues of harvested
 161 woody biomass. Revenues differ depending on the type of biomass and the downstream
 162 value chain in which it is used. Often, AFS biomass is used energetically, i.e. for direct
 163 heat or power generation, or as feedstock for anaerobic digestion typically creating
 164 low prices per tonne of biomass (Keegan et al., 2013). Alternatively, woody parts of
 165 the biomass (i.e., the stem and large branches) can also be used as pulp and paper
 166 grade material as well as feedstock for chemical industry or as timber achieving higher
 167 prices (Lamers et al., 2011). This price differentiation mirrors long-established forestry
 168 product classification systems. These classes range from veneer logs and sawlogs at the
 169 top, through pulpwood, to fuelwood and energy wood at the bottom (Sandberg et al.,
 170 2023). Wood categorization thereby depends primarily on stem diameter, straightness,
 171 and the presence of defects such as knots or curvature (Petráš et al., 2008). Large,
 172 defect-free stems can be directed to higher-value sawlog and veneer markets while
 173 smaller-diameter or lower-quality material is confined to pulp, particleboard, or energy
 174 uses. Thus, tree age at harvest is directly linked to achievable biomass revenues.
 175 While SRAFS typically supply the lower end of this quality cascade, LRAFS allow
 176 producing larger-diameter, higher-grade stems (Skog et al., 2014). Thus, fostering AFS
 177 establishment by maximizing economic returns from AFS biomass requires a strategy
 178 for allocating biomass fractions to the highest-value applications, while ensuring that
 179 the rotation length is aligned with the applications' quality requirements (Lamers et
 180 al., 2011).

181 Thus, for making educated decisions on AFS establishment and harvesting, the expected
 182 biomass quantities at harvest for stem, branches, and residues need to be estimated.
 183 In particular, the shares of the biomass types change with stand age. For example, the
 184 share of stem weight rises from roughly 55% at age 4 to over 75% at age 10 (Huber et
 185 al., 2018; Testa et al., 2014). Therefore, in [section 2](#) growth modeling approaches are

186 reviewed that capture the age-dependent growth dynamics of different biomass types
187 in AFS.

188 **Biomass growth models**

189 Individual-tree biomass is commonly approximated by species-specific allometric
190 power laws relating stem base diameter to tree biomass. Such relationships have been
191 calibrated e.g. for Central European poplar clones under SRAFS conditions (Huber et
192 al., 2016; Huber et al., 2018). For supply chain planning, however, stand-level biomass
193 per hectare depending on tree age is required. In the following, temporal biomass
194 growth models for different biomass fractions are derived. Therefore, a set of variables
195 is used which are summarized in [Table 1](#) in the appendix.

196 Stand-level biomass per hectare is denoted by $M(t)$. Stand-level biomass growth can
197 be described by different growth functions like the logistic, Gompertz, von Bertalanffy,
198 and Richards models which are also applied for other growth processes in biology
199 García (1993). All these functions show a right-skewed sigmoidal pattern and their
200 shape is controlled by a set of parameters (E. Tjørve & Tjørve, 2010). Models with more
201 parameters usually imply a more flexible growth model, but require more calibration
202 efforts which have to be counterweighted with modeling accuracy Fekedulegn et al.
203 (1999). Empirical studies indicate that modeling accuracy of parsimonious and more
204 extensive growth models are very similar in most cases Paula et al. (2013). Therefore,
205 in the following, a parsimonious 3-parameter Gompertz model (K. M. C. Tjørve &
206 Tjørve, 2017) is chosen for modeling $M(t)$, which is defined as

$$M(t) = A(N) \cdot \exp(-\exp(-k \cdot (t - t_0))). \quad (1)$$

207 Here, t_0 is the age of maximum instantaneous biomass growth and k determines
208 how quickly the stand approaches its site-specific asymptotic biomass quantity $A(N)$

209 (i.e. the maximum stand-level biomass quantity for $t \rightarrow \infty$). For Central European
 210 poplar in a medium-rotation AFS (10–15 yr), analysing empirical data from Huber et
 211 al. (2018), Niemczyk & Thomas (2021), and Jha (2018) yields parameter intervals of
 212 $k \in [0.17, 0.2]$ and $t_0 \in [9, 10]$ years. Parameter t_0 is often empirically estimated by
 213 analyzing current annual increments (CAI). Therefore, the stem base diameter of a
 214 sample of trees in a stand is measured in yearly rhythm and allometric laws are used
 215 to estimate tree-level biomass. This parameter indicates an orientation for the optimal
 216 rotation age as tree growth slows for higher ages.

217 The asymptote biomass quantity $A(N)$ depends on multiple factors, particularly on
 218 planting density and soil quality. It can be modeled by (Kira et al., 1961):

$$A(N) = C_{\text{site}} \cdot N^{1-\beta}, \quad (2)$$

219 where C_{site} encodes a site’s soil quality indicator, N indicates tree density (e.g. trees
 220 per hectare), and β is the competition exponent. Equivalently, the per-tree asymptote
 221 $A_{\text{tree}}(N) = A(N)/N = C_{\text{site}} \cdot N^{-\beta}$ decreases with density as individual trees have
 222 access to less growing space and more nutrition competition. Parameters C_{site} and
 223 β are highly site- and species-specific. For Poplar clones under European conditions,
 224 typically $\beta \in [0.3, 0.5]$ and $C_{\text{site}} \in [4, 6.5]$ (Niemczyk & Thomas, 2021). [Figure 10](#) in the
 225 appendix illustrates both asymptote functions for advantageous and disadvantageous
 226 site conditions.

227 When assessing potential downstream applications of AFS biomass, biomass fractions
 228 are to be defined meeting the requirements of downstream consumers Civitarese et
 229 al. (2019). For example, for timber and biochemical applications, stem diameters
 230 of AFS trees need to exceed a minimum threshold in order to qualify as a suitable
 231 feedstock (Zhu, 2011). In the following, three biomass fractions are distinguished based
 232 on common forestry standards: stems, branch wood, and residue. The stem fraction

consists of stems with a diameter of at least 15 cm, while branch wood requires a diameter of at least 7 cm. The residue fraction comprises smaller parts of the tree. To subdivide the total AFS biomass modeled by (1) into the three fractions depending on tree age, a logistic growth model (Zeide, 1993) is used:

$$q_p(t) = \frac{f_p}{1 + e^{-r_p \cdot (t - t_{50,p})}}, \quad (3)$$

where f_p denotes the asymptotic share of fraction p for mature trees. Parameter $t_{50,p}$ indicates the inflection point and coincides with the age at which 50 % of the asymptotic share of fraction p is reached. Using empirical data Niemczyk & Thomas (2021), $r_{stem} = 0.467, \text{yr}^{-1}$, $t_{50,stem} = 8.19$ years, and $f_{stem} = 0.75$ is estimated for the stem fraction. For the branch fraction, parameters $r_{bran} = 0.698 \text{ yr}^{-1}$, $t_{50,bran} = 7.55, \text{yr}$, and $f_{bran} = 0.15$ can be assumed. Figure 9 in the appendix illustrates the development of the fraction shares over stand age. The share of the residue fraction is implicitly defined as $q_{resid}(t) = 1 - q_{stem}(t) - q_{branch}(t)$.

Combining equations (1)–(3), the dry biomass yield of fraction p per hectare at harvest age t is:

$$\eta_p(t) = q_p(t) \cdot M(t). \quad (4)$$

Fresh-weight yields at felling are obtained from dry matter yields via $\eta_p^{\text{fw}}(t) = \eta_p(t)/(1 - \omega)$, where the moisture content at felling $\omega = 0.55$ (Civitarese et al., 2019). Figure 1 shows the fractions' stacked biomass quantities $\eta_p(t)$ depending on stand age compared with empirical observations from Jha (2018) and Huber et al. (2018).

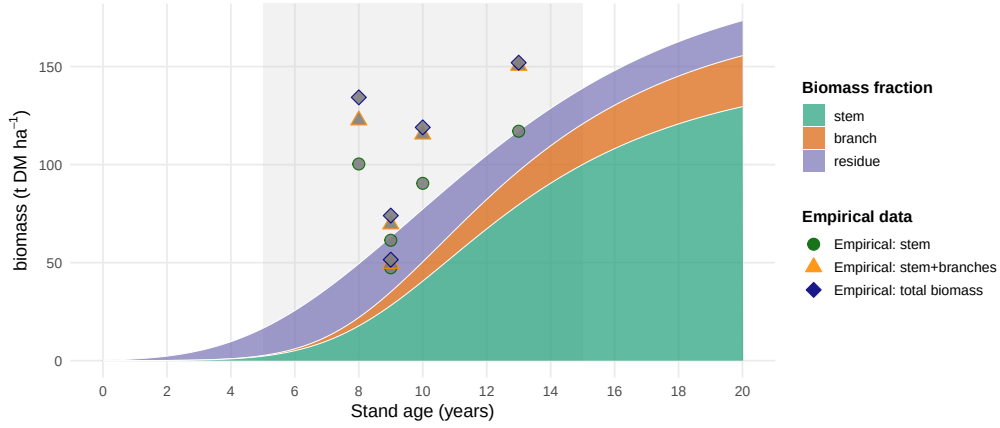


Figure 1: Stand-level aboveground biomass components for poplar under AFS management with $N = 250$ trees, ha^{-1} . Grey band: typical rotation times (5-15 years).

Biomass supply chain design

The design and planning of supply chains based on industrial biomass feedstocks has attracted substantial research attention over the past decades, driven by the need to ensure cost-efficient and sustainable feedstock provision for bioenergy and biorefinery applications (see e.g. extensive reviews from Gital & Bilgen (2024), Zahraee et al. (2020), and De Meyer et al. (2014)). Despite its name, the term biomass supply chain (BSC) is often associated with exclusive energetic uses of biomass and often equivalent to biorefinery supply chain (see e.g. Espinoza et al. (2017)). The considered biomass feedstocks range from agricultural energy crops, over urban biomass residues to forestry wood (see e.g. Sharma et al. (2013) for an overview). Literature on forestry supply chains (FSC) is the oldest sub-branch in this literature stream (Cambero & Sowlati, 2014). Here, the focus is on one particular biomass feedstock, but usually multiple fields of applications are considered (see e.g. D'Amours et al. (2008) for an overview of fields of applications). In general, the structure of BSC and FSC typically encompasses three process types: i) biomass production, ii) harvesting and logistics, and iii) biomass processing. These processes are modeled at various levels of details depending on the

267 scope of decisions to be supported. The decision scopes can be distinguished in: (i)
 268 strategic network design (e.g. facility location, species selection), (ii) tactical planning
 269 (e.g. harvest scheduling, inventory capacities), and (iii) operations (e.g. routing of
 270 vehicles, inventory control) (De Meyer et al., 2014). This manuscript is dedicated
 271 to agroforestry supply chain design. That is, the focus is put on a specific type of
 272 biomass and multiple fields of application. In the following relevant literature from FSC
 273 and BSC literature is reviewed which contributes to this type of problem. Relevant
 274 aspects from BSC literature mainly involve biomass production economics and strategic
 275 design modeling while relevant aspects from FSC literature is concentrated on wood
 276 logistics and industrial applications. Note that there is no existing literature specifically
 277 dedicated on agroforestry supply chain design to the best of the author’s knowledge.

278 One of the earliest FSC models is proposed by Gunnarsson et al. (2004). Here, a cost-
 279 optimal distribution and processing network for supplying heating plants with wood
 280 chips is to be found. Therefore, a multi-period MILP model is formulated satisfying
 281 feedstock demand at heating plants by selecting harvesting areas and deciding in each
 282 period (i.e., each year) about chipping, storing, and transporting forest biomass (like
 283 logs). Eksioglu et al. (2009) develop one of the earliest MILP models for biomass-
 284 to-biorefinery supply chain design. The focus is on biomass conversion processes
 285 by integrating facility location, capacity sizing, and transport mode selection for
 286 lignocellulosic feedstocks. The model minimizes total system cost assuming, that
 287 biomass supply is an exogenous input, i.e. biomass production is not modeled explicitly.

288 De Meyer et al. (2016) proposes a similar profit-maximizing MILP for strategic and
 289 tactical decision support. In this model, the focus is on pre-treatment and storage
 290 processes of different types of grassland biomass types (grass, hay, straw, etc.) supplying
 291 anaerobic digesters for energy generation. Walther et al. (2012) describe wood-based
 292 biomass supply chain for biofuel production where the biomass is converted either
 293 directly into biofuels or indirectly via an intermediate material (slurry). The objective

294 is here maximize the net present value (NPV) over the planning horizon discounting
295 the yearly cash flows of the supply chain. The revenue is created by supplying refineries
296 with the supply chain’s biofuels. Similar to Gunnarsson et al. (2004), the refineries’
297 biofuel demand can be satisfied by imports, such that a possible solution is that no
298 regional biomass is used to produce biofuel. The aforementioned models consider only
299 biomass-to-energy supply chains.

300 Mansoornejad et al. (2013) extend this perspective by integrating product and process
301 portfolios. Their model considers multiple biomass conversion processes as well as
302 sources of lignocellulosic biomass. It accounts for product conversion efficiencies along
303 different processing pathways and maximizes profit over a multi-period time horizon.
304 Likewise, Cambero et al. (2015) model energetic and chemical conversion pathways
305 of lignocellulosic biomass in a strategic supply chain design model for maximizing
306 NPV over a 20-year period. García-Velásquez et al. (2022) propose a bi-objective
307 mixed integer linear program (MILP) for designing a supply chain for producing
308 Polyethylene Terephthalate (PET) based on wheat and sugar beets. In this model, both
309 biomass types are required to produce intermediate chemicals (mono ethylene glycol
310 and purified terephthalic acid) which are merged into PET such that a 4-stage supply
311 chain is modeled. For each processing step locations have to be selected among a set of
312 candidate locations. The objectives pursued involve total supply chain costs and GHG
313 emissions. The trade-off between cost and GHG emissions is studied by incorporating
314 GHG cost and varying GHG prices. A similar 4-stage bio-chemical network design
315 problem for biobutanol production based on microalgae is studied in Arabi et al. (2019).
316 Recently, Roudneshin & Sosa (2025) presents a novel procedure for a bioenergy supply
317 chain design problem by combining mathematical programming with an Analytic
318 Hierarchy Process (AHP). The AHP pre-processes potential candidate locations for
319 biorefineries where biomass wastes are converted into biogas which is subsequently fed
320 into the local gas grid.

321 None of the reviewed articles from BSC and FSC literature considers biomass pro-
 322 duction processes explicitly, but model biomass feedstock as an exogenous resource.
 323 This is because most approaches concentrate on applications where waste biomass is
 324 considered as feedstock. In FSC applications, too, production decisions are usually
 325 not part of supply chain design decisions as growing cycles in forest management are
 326 even longer than the time horizons in supply chain design problems. Both aspects are
 327 substantially different in AFS. Therefore, this manuscript contributes to literature
 328 by proposing a novel agroforestry supply chain design problem (AFS-SCD) which
 329 explicitly models biomass production as an integral part of the supply chain design
 330 problem. In agroforestry applications, biomass production decisions depend on the
 331 existence of downstream markets for the produced wood. Moreover, multiple types
 332 of biomass and multiple application industries are considered. In the following, the
 333 AFS-SCD is outlined in detail.

334 **MILP Model for Agroforestry Supply Chain Design**

335 This section describes the AFS supply chain design problem (AFS-SCD), a lin-
 336 ear programming model for the integrated design and operation of an agroforestry
 337 biomass supply chain with multiple AFS biomass types used as feedstocks for potential
 338 consumers in chemical, pulp and paper as well as energy industry.

339 **Sets, Indices, and Time Structure**

340 The model considers a set of agroforestry sites $\mathcal{I}$, storage/pre-treatment facilities
 341 $\mathcal{J}$, consumer sites $\mathcal{K}$, and biomass types $\mathcal{P} = \{stem, branches, residues\}$, whose
 342 growth models are outlined above. Time is discretized into annual periods $t \in \mathcal{T} =$
 343 $\{1, \dots, T_{\max}\}$. To reduce complexity and discard implausible options, $A^{\min}$ and $A^{\max}$
 344 define minimum and maximum stand ages. Thus, AFS harvests are possible only
 345 in periods $\mathcal{T}^{\text{harv}} = \{A^{\min} + 1, \dots, T_{\max}\}$. To model AFS establishment and harvest

346 decisions, the following arc sets are defined: $\mathcal{S}^{est}$ defines potential establishments
 347 of AFS. If an AFS is established in period $t \in \mathcal{T}$, arc $(0, t)$ is selected. Harvest set
 348 $\mathcal{S}^{harv}$ summarizes all potential consecutive harvest periods within the age interval
 349 $[A^{\min}, A^{\max}]$. That is, if an AFS is established or harvested in period s and the next
 350 harvest is in period t , arc (s, t) is selected. Finally, if there is no subsequent harvest
 351 after period t , the AFS is terminated by selecting arc $(t, T^{\max} + 1)$. Termination arcs
 352 are summarized in set $\mathcal{S}^{term}$. All arc sets are formally defined as follows:

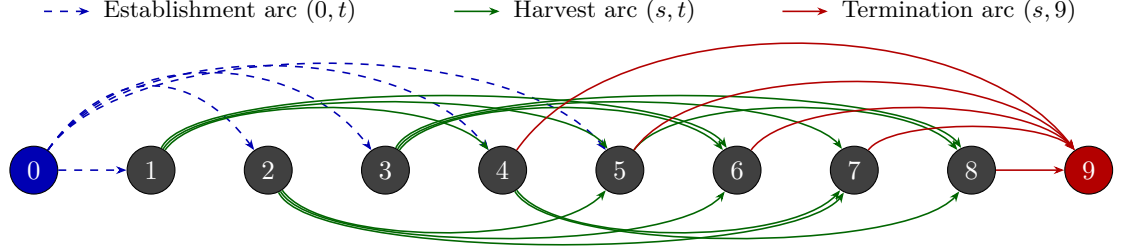
$$\begin{aligned}\mathcal{S}^{est} &= \{(0, t) \mid t \in \{1, \dots, T^{\max} - A^{\min}\}\}, \\ \mathcal{S}^{harv} &= \{(s, t) \mid s, t \in \mathcal{T} \wedge t - s \in \{A^{\min}, \dots, A^{\max}\}\}, \\ \mathcal{S}^{term} &= \{(s, T^{\max} + 1) \mid s \in \{A^{\min} + 1, \dots, T_{\max}\}\}, \\ \mathcal{S} &= \mathcal{S}^{est} \cup \mathcal{S}^{harv} \cup \mathcal{S}^{term}.\end{aligned}$$

353 [Figure 2a](#) illustrates all arc sets for $T = 8$, $A^{\min} = 3$, $A^{\max} = 5$.

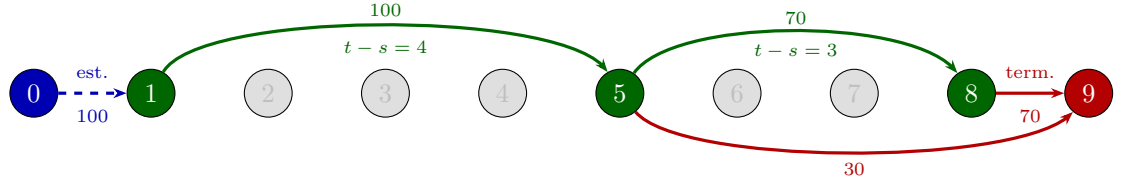
354 Decision Variables

355 Temporal flow variables z_{ist} indicate which quantity of the area of site i is dedicated
 356 to AFS operations in the period $(s, t) \in \mathcal{S}$. Depending on the arc type, this implies
 357 establishment, harvest, or termination. Selected arcs build a network flow over the
 358 complete planning horizon from $t = 0$ to $T^{\max} + 1$. [Figure 2b](#) illustrates a simple
 359 network flow with establishment of an AFS in period 1 and harvests in periods 5 and 8.

360 Spatial flow variables X_{ijpt} and $X_{jkpp't}$ represent transport of products from AFS sites
 361 to storage/pre-treatment facilities and from there to consumers, respectively. Variable
 362 S_{jpt} captures end-of-period inventory at facility j and reflects yearly storage quantities.
 363 [Table 2](#) in the Appendix summarizes all sets, decision variables, and parameters used
 364 in the MILP formulation, while η_{pa} refers to (4).



(a) Complete sets of arcs $\mathcal{S}^{est}$, $\mathcal{S}^{harv}$, and $\mathcal{S}^{term}$.



(b) Example of a feasible path through the network with establishment in period 1, harvests in periods 5 and 8, termination in period 9.

Figure 2: Network structure of set $\mathcal{S}$ for $T = 8$, $A_{\min} = 3$, $A_{\max} = 5$. Dashed blue arcs: establishment arcs $\mathcal{S}^{est}$, solid green arcs: harvest arcs $\mathcal{S}^{harv}$, solid red arcs: termination arcs $\mathcal{S}^{term}$.

Objective Function

As discussed in [section 2](#), the economic attractiveness of AFS supply chains is modeled from a supply chain perspective taking into account all processes, activities, and participants involved in finally serving consumer markets with AFS-based feedstocks. Therefore, revenues created by selling AFS-based feedstocks to different industrial consumers are counterweighted by establishment, opportunity, harvest, transport, and storage costs of landowners, agricultural firms as well as biomass logistics companies. The model maximizes total profit of all participants in the AFS supply chain over the planning horizon as follows:

$$\max Z = \sum_{k \in \mathcal{K}} \sum_{p \in \mathcal{P}} R_{kp} \cdot \sum_{j \in \mathcal{J}} \sum_{(p', p) \in \mathcal{Q}} \sum_{t \in \mathcal{T}} X_{jkp'pt} \quad (5)$$

$$\begin{aligned}
& - \sum_{i \in \mathcal{I}} c_i^{est} \cdot \sum_{(0,t) \in \mathcal{S}^{est}} z_{i0t} \\
& - \sum_{i \in \mathcal{I}} (c_i^{main} + c_i^{opp}) \cdot (t - s) \cdot \sum_{(s,t) \in \mathcal{S}^{harv}} z_{ist} \\
& - \sum_{i \in \mathcal{I}} c_i^{harv} \cdot \sum_{(s,t) \in \mathcal{S}^{harv}} z_{ist} \\
& - \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} \sum_{p \in \mathcal{P}} \sum_{t \in \mathcal{T}} c_p^{tr-raw} \cdot d_{ij} \cdot X_{ijpt} \\
& - \sum_{j \in \mathcal{J}} \sum_{k \in \mathcal{K}} \sum_{(p,p') \in \mathcal{Q}} \sum_{t \in \mathcal{T}} c_{p'}^{tr-pre} \cdot d_{jk} \cdot X_{jkpp't} \\
& - \sum_{j \in \mathcal{J}} \sum_{p \in \mathcal{P}} \sum_{t \in \mathcal{T}} c_j^{stor} \cdot S_{jpt}
\end{aligned}$$

Here, R_{kp} are product- and consumer-specific revenues for each ton of biomass fraction p delivered to consumer k (in €/t). The opportunity cost term penalizes the land use when AFS are established. Specifically, c_i^{opp} can be negative or positive depending on the site-specific conditions. It reflects effects of lost field crop revenues due to AFS as well as higher yields of remaining field crop areas (e.g. due to wind and water erosion protection by AFS). While the assessment of lost field crop revenues is usually well predictable, the economic evaluation of positive ecological effects of AFS is more difficult. Thus, c_i^{opp} may also contain regulatory subsidies to support establishing and maintaining AFS. Transport cost terms mirror the two-stage flow structure of the AFS supply chain with differentiated transport cost rates for raw and pre-treated biomass depending on the biomass type. Cost rates differ because, for example, density and water content of the shipped biomass types differ or because specialized equipment (e.g. for log transports) is used (Väättäinen et al., 2021). Moreover, site-specific establishment cost for planting the AFS trees is considered, next to yearly maintenance and harvesting costs. All these costs scale with the size of a site dedicated to AFS operation z_{ist} measured in hectares. Storage costs are considered if biomass

types are stored for more than a year. Storage cost rate c_j^{stor} measures the costs for storing a ton of biomass for one year and comprises direct storage cost (e.g. for storage site maintenance) as well as opportunity cost due to potential material losses and capital commitment. AFS biomass harvested and consumed in the same year incur no storage cost. Usually AFS are harvested at the beginning of a year and usually dries during spring and summer before used in downstream industries. As under-year drying is standard for most industrial applications and beneficial for transport economics, drying storage in the year of harvest is not associated with additional cost in this study.

Constraints

Constraints (6) limits the area dedicated to AFS operations to the maximum size of site i . Constraints (7) enforce flow conservation in each period. If AFS systems of size z_{ist} are established or harvested in period $t \in \mathcal{T}$, there needs to be a succeeding activity (harvest or termination) of this area. This way, (7) guarantee valid (possibly empty) AFS activities over the complete time horizon for each site. Constraints (8) link the harvest-cycle variables z_{ist} to physical yield quantities $\eta_{p(t-s)}^{fw}$ for age $t - s$ as defined above. Thus, constant yields per hectare are assumed for each site i . Note that this simplification can be dropped by introducing site-specific yields $\eta_{ip(t-s)}^{fw}$ without increasing numerical complexity. Moreover, harvest quantities can be smaller than available AFS biomass implying that unharvested biomass quantities are lost (i.e., no revenues are generated). Constraints (9) link harvest quantities to transport flows and ensure inventory balance at each facility. Biomass losses due to drying ($\Delta \leq 1$) and biomass conversion processes ($\delta_{pp'} \geq 1$) are considered. Constraints (10)–(11) enforce storage and processing capacity limits at each facility j . Constraints (12) implement product usage at multiple consumption sites k . Here, $(p', p) \in \mathcal{Q}$ defines a product mapping allowing high-quality product p' (e.g. stems) satisfying demand of lower-quality products p up to maximum demand $D_{kpt}^{\max}$.

$$\sum_{(0,t) \in \mathcal{S}^{est}} z_{i0t} \leq AREA_i \quad \forall i \in \mathcal{I} \quad (6)$$

$$\sum_{(s,t) \in \mathcal{S}} z_{ist} = \sum_{(t,u) \in \mathcal{S}} z_{itu} \quad \forall i \in \mathcal{I}, t \in \mathcal{T} \quad (7)$$

$$\sum_{j \in \mathcal{J}} X_{ijpt} \leq \sum_{(s,t) \in \mathcal{S}^{harv}} \eta_{p(t-s)}^{fw} \cdot z_{ist} \quad \forall i \in \mathcal{I}, p \in \mathcal{P}, t \in \mathcal{T}^{harv} \quad (8)$$

$$S_{jpt} = \Delta \cdot S_{jp,t-1} + \sum_{i \in \mathcal{I}} X_{ijpt} - \sum_{k \in \mathcal{K}, (p,p') \in \mathcal{Q}} \delta_{pp'} \cdot X_{jkpp't} \quad \forall j \in \mathcal{J}, p \in \mathcal{P}, t \in \mathcal{T}^{harv} \quad (9)$$

$$\sum_{p \in \mathcal{P}} S_{jpt} \leq CAP_j^{stor} \quad \forall j \in \mathcal{J}, t \in \mathcal{T} \quad (10)$$

$$\sum_{i \in \mathcal{I}, p \in \mathcal{P}} X_{ijpt} \leq CAP_j^{proc} \quad \forall j \in \mathcal{J}, t \in \mathcal{T}^{harv} \quad (11)$$

$$\sum_{j \in \mathcal{J}, (p',p) \in \mathcal{Q}} X_{jkp'pt} \leq D_{kpt}^{\max} \quad \forall k \in \mathcal{K}, p \in \mathcal{P}, t \in \mathcal{T}^{harv} \quad (12)$$

$$z_{ist}, X_{ijpt}, X_{jkp'pt}, S_{jpt} \geq 0 \quad \forall i, j, k, p, p', t \quad (13)$$

416 In summary, AFS-SCD constitutes an LP which allows to decide on which fraction of
 417 a site, AFS are established and how the AFS operations are scheduled in a multi-year
 418 planning horizon. The AFS-SCD takes a supply chain perspective and maximizes total
 419 profit of all actors involved in these processes. Thus, it does not explain how the profits
 420 are split e.g. between farmers, AFS service providers and logistics service providers.
 421 The continuous modeling of AFS areas in z_{ist} leads to a comparatively easy to solve
 422 problem.

Case Study

The AFS-SCD is intended to support the design of regional agroforestry biomass supply chains fostering regional bioeconomy clusters connecting multiple industrial sectors. The proposed AFS-SCD is instantiated on a real-world biomass supply chain case study located in the tri-state region of Saxony-Anhalt, northern Thuringia, and northern Saxony, Germany. This region is characterized by a high density of agricultural land usage as well as cluster of paper, wood, and chemical industry. The regional chemical industry relies on fossil resources so far, but is under substantial transformative pressure which led to investments in sustainable bio-based production assets requiring sustainable regional feedstocks. [Figure 11](#) in the Appendix shows the considered agricultural production sites along with the set of potential wood-processing hubs, and the regional users of AFS biomasses from chemical, paper, and wood-processing industry as well as the energy sector.

The supply chain comprises 2239 agricultural production parcels which are grouped into $|\mathcal{I}| = 145$ potential AFS production sites. Clustering is based on geographical distances between parcel centroids using a k -means approach because it is unclear which agricultural firms manage which parcels. Pre-treatment and intermediate storage facilities $\mathcal{J}$ are located on existing wood processing facilities as well as potential new log or wood chip yards. All harvested biomass quantities at AFS sites are transported to logistics hubs first. At the hubs sorting and transshipment processes take place before biomass types are forwarded to the consumer facilities. Storing and transshipment capacities at the hubs are assumed to be constant in time and universally set to 40909 kt per year. Finally, $|\mathcal{K}| = 28$ consumer sites are included. Each consumer site belongs to an existing facility from chemical, paper/wood processing industry, or the energy sector. Distances between the facilities d_{ij} and d_{jk} are calculated as road distances between the facilities' coordinates. For each consumer facility, an yearly upper demand

capacity for each of three biomass product types are given along with consumer-specific prices. The following biomass types are distinguished:

- **P1: stem wood – chemical/paper grade:** long-fibre wood chips from stems suitable for conversion in a local biorefinery and paper factory. In total, a yearly demand of 620 kt and an average prices of 45 € is assumed.
- **P2: branch wood – sawmill/pulp grade:** medium/short-fibre wood chips from larger branches suitable for mechanical processing in sawmills and lower-grade pulping. Demand is distributed across 6 smaller facilities and accumulates to 410 kt/year. Gate prices range from 28-38 €/t.
- **P3: residue – energy grade:** residual wood fractions (chips, bark, off-cuts) used for heat and electricity generation or lower-grade pellet production. Demand accumulates among 24 consumers to 132 kt/year. Gate prices range from 18-24 €/t.

It is assumed that the biomass product types show a quality cascade which means that the demand of pulp grade (P2) can also be satisfied by chemical/paper grade biomass (P1). Likewise, energy grade demand (P3) can be satisfied by all biomass types. Note that constant demands over the planning horizon are assumed, although an increasing biomass demand can be expected due to the green transition in those industries. From this perspective, a rather conservative market demand scenario is assumed.

The growth model introduced in [section 2](#) is used to model biomass quantities depending on age and the area dedicated to AFS. The growth model is parameterized for poplar clones at central European conditions as introduced above with a planting density of $N = 400$ trees per hectare. If AFS are established, they have to be harvested within $A^{min} = 3$ to $A^{max} = 20$ years allowing a wide range of potential harvesting strategies between short-rotation and long-rotation AFS. As the total planning horizon is set to $|\mathcal{T}| = 40$ years, multiple harvesting cycles can be established. The sites do not differ with respect to growth conditions as the climatic and soil quality variation within the

comparatively small region is rather limited. However, the sites differ with respect to size (given in hectares) and opportunity cost rates (given in € per hectare and year). Because site-specific opportunity cost rates c_i^{opp} are very difficult to determine and to study the effect of this parameter on site selection, this parameter is sampled based on a normal distribution with mean 521 and standard deviation 391. For most sites, substantial opportunity costs are assumed reflecting the loss in field-crop revenue, risk premiums of landowners as well as higher costs for field crop maintenance. However, 0 sites with negative opportunity cost rates reflect sites where AFS lead to an increase in field crop productivity on the remaining field area. By varying opportunity cost rates, the effect of opportunity cost on site selection for AFS can be studied which gives valuable insights for communication with farmers and politics. Subsidies for establishing AFS are one of the core instruments to foster their adoption. The height and effectiveness of those subsidies, however, is controversially disputed. Remaining cost parameters are universally set for all sites and comprise maintenance cost rates c_i^{main} (€ per year and hectare), establishment cost c_i^{est} (€ per hectare) and harvesting cost c_i^{harv} (€ per hectare). Likewise, transport cost rates for harvested biomasses c_p^{tr-raw} and pre-treated/sorted biomasses $c_{p'}^{tr-raw}$ are set universally. All scalar parameters are shown in [Table 3](#) in the Appendix.

Optimization results

The AFS-SCD model is solved for the instance data described above with Gurobi. The model selects a subset of sites for AFS establishment and identifies the optimal harvest schedules, intermediate facility utilization, and product allocation across the planning horizon. [Figure 3](#) shows the spatial configuration of the profit-optimal supply chain configuration.

In total, on 38 sites AFS are established which corresponds to about 26 % of all sites and 24 % of the total farm area. Over the planning horizon, 26368 kt of AFS biomass

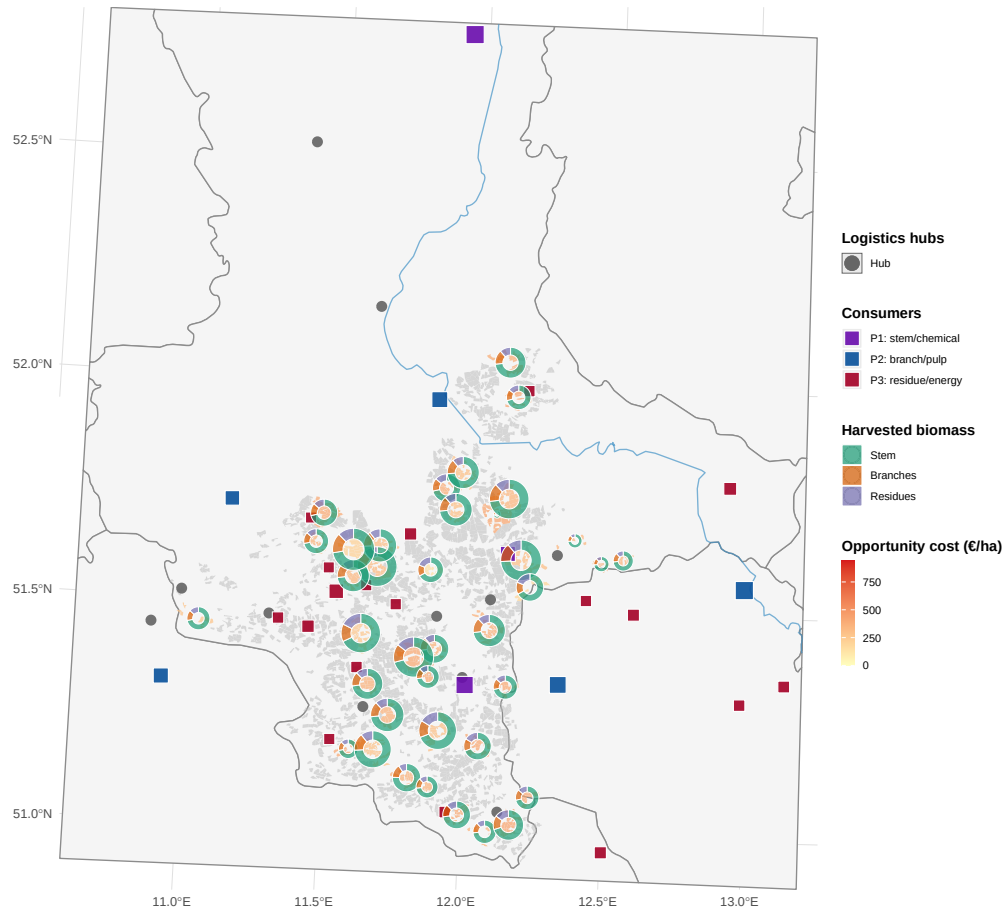


Figure 3: Site selection and biomass production in optimal supply chain configuration. Green polygons: active sites with biomass harvest; grey polygons: non-selected sites. Pie charts at active sites show the product mix of harvested biomass, scaled by total harvested volume.

501 is produced on these sites which suffices to satisfy 57 % of total feedstock demand.

502 [Figure 4](#) shows the harvested biomass quantities over time along with total maximum
503 demand for each biomass type.

504 During the first 9 years, no AFS biomass is harvested as trees on established AFS need
505 to grow up to a certain size before first harvest. Thus, about 25,% of consumers' total
506 biomass demand cannot be satisfied due to this grow-up period. Afterwards, a cyclical
507 pattern appears with a harvesting peak between year 17–19 and a breakdown in year

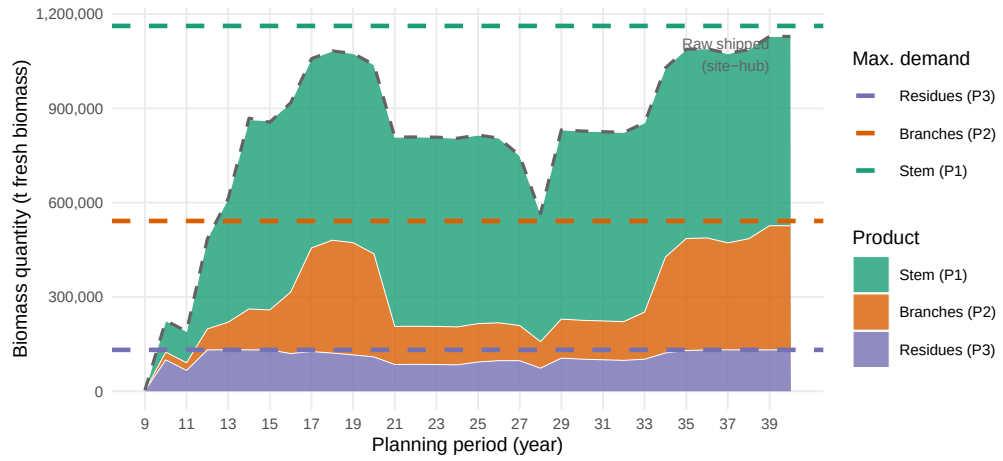


Figure 4: Total biomass delivered to consumers by product type over the planning horizon. Stacked areas show period-level volumes of stem (P1), branches (P2), and residues (P3) flowing from storage hubs to all consumers. Dashed line: total harvested raw biomass shipped from sites to hubs.

28 before a plateau is reached from year 35-40. The harvesting quantities of stems and
 509 branches show a similar pattern, while quantities of harvested energy grade biomass
 510 are rather stable and close to the demand limit over the planning horizon. Thus,
 511 rather long-rotation AFS are established as Figure 5 confirms showing the histogram
 512 of harvesting ages across all sites.

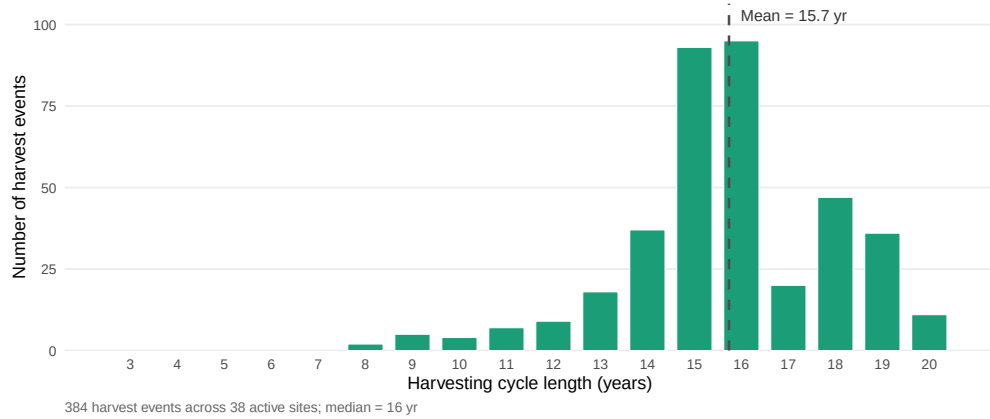


Figure 5: Distribution of harvesting cycle lengths at active agroforestry sites. Each bar represents the number of individual harvest events (over all periods and active sites) whose cycle length (stand age at harvest, i.e. $t - s$) falls in that year bin.

513 The histogram illustrates that the majority of harvest events occurs at stand ages
 514 between 15–16 years, indicating that long-rotation AFS seem to be more profitable in
 515 this setting. Short-rotation cycles (<10 years) occur only in marginal frequencies. This
 516 optimal rotation length might be caused either due to the limited demand of energy-
 517 grade biomass or the price gaps between biomass types or a combination of both effects.
 518 The quantity of energy-grade biomass created in long-rotation AFS suffices to cover
 519 total energy-grade biomass demand almost completely. Rotation length is focused on
 520 production of chemical-grade biomass where highest prices are assumed. Focusing
 521 on chemical industry supply requires a production of trees with comparatively large
 522 stems. This strategy maximizes revenue, but in the same time increases opportunity
 523 costs as wood harvests appear in larger time intervals. Figure 6 presents the aggregate
 524 cost–profit waterfall over all periods and actors (left side) as well as cash flow, cost,
 525 and revenue development over time (right side).

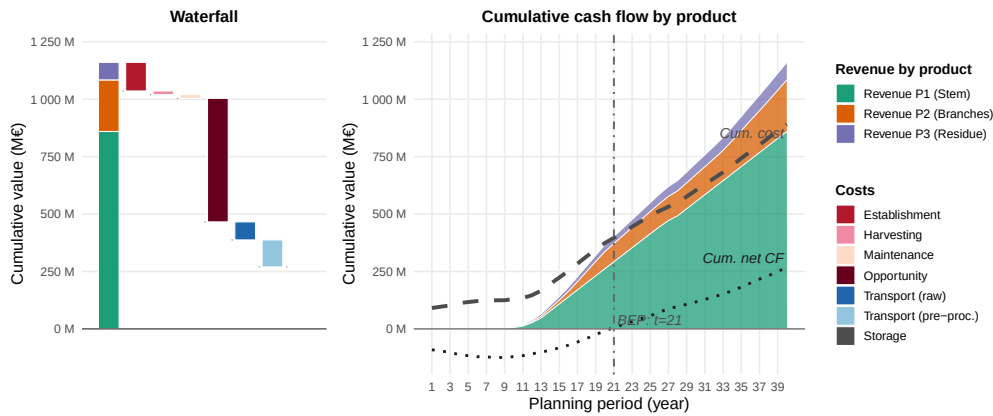


Figure 6: Total cost–profit breakdown across all actors and years. The left panel (waterfall) decomposes the objective value into revenue and individual cost categories; the right panel shows the period-level trajectory of revenues and stacked costs.

526 It appears that revenue is dominated by selling chemical-grade biomass. The domi-
 527 nating cost component is opportunity cost although active sites are those sites with a
 528 comparatively small opportunity cost rate. Next to opportunity cost, transport and

establishment costs are the most relevant cost drivers highlighting the importance of transport distances for profitability. Analyzing total cash flow over time shows that break-even is reached after about 20 years, i.e. after the first harvests have occurred on most AFS sites. Afterwards, cash flow is continuously increasing indicating that establishing AFS is a strategic change in agricultural practice and pays only if this strategy is implemented and maintained over comparatively long time spans. This underpins the importance of strategic buying contracts with industrial partners in order to reduce revenue risks of farmers when deciding to establish AFS systems.

Although profitability of the total AFS supply chain is given for the case study data, profitability of AFS operations on site level are more heterogeneous. Figure 7 shows revenues and costs split by component and normalized to AFS size and active years of AFS. Additionally, the net profit in € per year and hectare is shown.

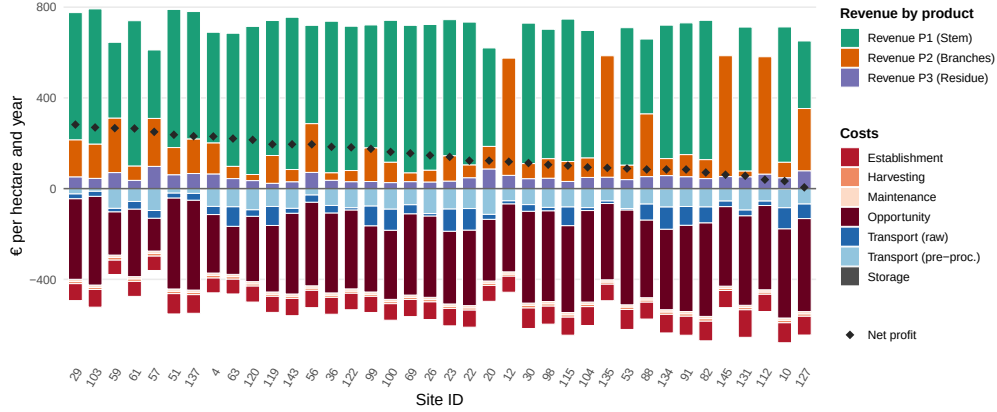


Figure 7: Revenues, costs and net profit for sites with AFS operations split by type and expressed in € per hectare per year of AFS operation.

It appears that net profitability of AFS ranges between 6 and 282 € per hectare and active year. Note that this value constitutes additional profit after subtracting operational cost as well as opportunity cost of lost profit from field crop. That is, for calculating the realized net margin, opportunity cost can be added such that realized

yearly net margin ranges between 210 and 473 € per hectare. Still, relative net profit is comparatively low compared to ordinary agricultural practices with profitable field crops (like winter wheat). The revenue splits of active sites indicate that at some sites medium-grade and low-grade biomass types contribute substantially or even solely to total revenue (see e.g. site 12, 135, 145, and 112). This indicates that site-specific AFS strategies might differ from the overall supply chain strategy. In other words, strategic niches seem to exist w.r.t. the consumer segments supplied by a farm. This might also be reflected in the rotation length and, although not discussed here, the selection of trees and clones used in AFS. Moreover, profitability of AFS sites seems to depend on various factors like local opportunity costs as well as closeness to consumers. Figure 8 shows net margins and opportunity cost rates of sites with active AFS.

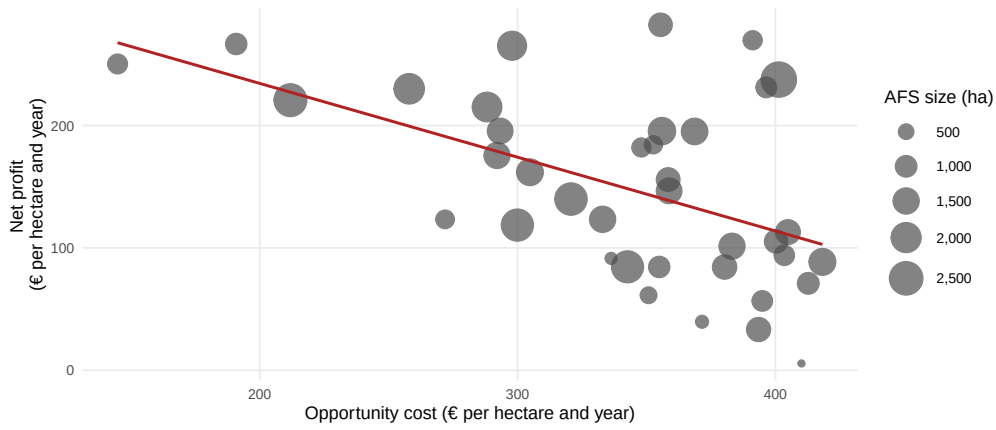


Figure 8: Scatterplot of opportunity cost rate and net margin (both in € per ha and year) for each site with active AFS. Size of AFS reflected in point size.

As expected, net profit of AFS depend heavily on a site's opportunity cost rate showing a clear negative trend. Moreover, it appears that AFS sizes vary quite heavily between 86 and 2794 hectare. It is likely that very small sites will not operate efficiently in practice, because costs for establishment, harvesting, and maintenance do not scale perfectly with size, but have a fixed cost component.

561 Conclusion

562 This paper proposes the first agroforestry supply chain design problem. To this end,
563 a linear program is formulated that integrates biomass production, processing, and
564 allocation decisions. The model endogenizes biomass production decisions using arc-
565 based age-lag variables. Arc weights are precomputed based on a Gompertz growth
566 model for biomass fractions like stem, branch, and residue. Biomass processing is
567 modeled in a 2-stage logistics network, where biomass fractions are consolidated at
568 pre-treatment and logistics hubs. Subsequently, biomass fractions are allocated to
569 different consumer facilities based on technical suitability and potential revenue. The
570 formulation is, to the best of the author’s knowledge, the first supply chain model to
571 simultaneously optimize AFS establishment decisions, consecutive harvest scheduling,
572 and allocation of biomass fractions across heterogeneous consumers.

573 The case study for the tri-state region in southern Saxony-Anhalt demonstrates the
574 practical relevance of the approach. The optimal supply chain establishes AFS on
575 only a subset of the available sites and area, concentrating establishment on sites
576 with comparatively low opportunity cost while sites with high opportunity cost are
577 rarely selected. Long-rotation AFS strategies dominate the optimal solution, as they
578 maximize the share of large-diameter stem biomass eligible for high-value chemical-
579 and pulp-grade markets. Correspondingly, revenue is dominated by chemical-grade
580 biomass sales, while opportunity cost, establishment, and transport costs are the most
581 relevant cost drivers. Nonetheless, also sites with only energy-grade biomass production
582 can be operated profitably. This finding implies that the establishment decision of
583 an AFS requires a clear market strategy which affects not only AFS sizing but also
584 species selection and harvest scheduling. Break-even at the aggregate supply chain
585 level is reached after about 20 years, once the first large-scale harvest occurs. This
586 finding highlights that AFS adoption constitutes a long-term strategic commitment
587 that depends on reliable off-take agreements with industrial partners. At the site

588 level, net profitability is highly heterogeneous and correlates negatively with local
589 opportunity cost. This indicates that AFS viability is strongly conditioned by regional
590 land-use competition.

591 Several limitations point to directions for future research. First, consumer demand is
592 exogenous. endogenizing demand responses to changing biomass cost would require
593 estimating an (inverse) demand function for each biomass fraction market. Second,
594 the case study uses a single growth model reflecting poplar plantations in a Central
595 European environment. Specifying multiple growth models for multiple tree species
596 and clones would allow to integrate tree and clone selection decisions into the model,
597 provided growth conditions are site-specific and can be estimated. Such an feature
598 would also help to streamline market strategies for each AFS site. Third, environmental
599 co-benefits of AFS such as carbon sequestration, nitrate leaching reduction, and
600 biodiversity enhancement are not considered in the current model. Integrating life
601 cycle assessment indicators as additional objectives or constraints would support a
602 more comprehensive sustainability evaluation of AFS-based supply chains. Fourth, the
603 model can be extended to stochastic settings by introducing stochastic yield coefficients
604 leading to stochastic or robust optimization approaches. Such an extension would help
605 deciders to quantify economic and environmental resilience of AFS.

606 **Author contributions**

607 The author is responsible for all contents of this text.

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619 and analysis, decision to publish, or preparation of the manuscript.

620 **Conflict of interest**

621 The author declares no potential conflict of interests.

622 **Data availability statement**

623 The interactive web application used to parameterise, run, and explore the AFS-SCD
624 model developed in this study, together with the underlying case study data, is openly
625 accessible at https://tk-mlu-pl.shinyapps.io/DIP_SmartAgroforst_SCD/. All figures
626 and results reported in this manuscript can be reproduced through this application.

627 **Supporting information**

628 Additional supporting information may be found in the online version of the article at
629 the publisher’s website.

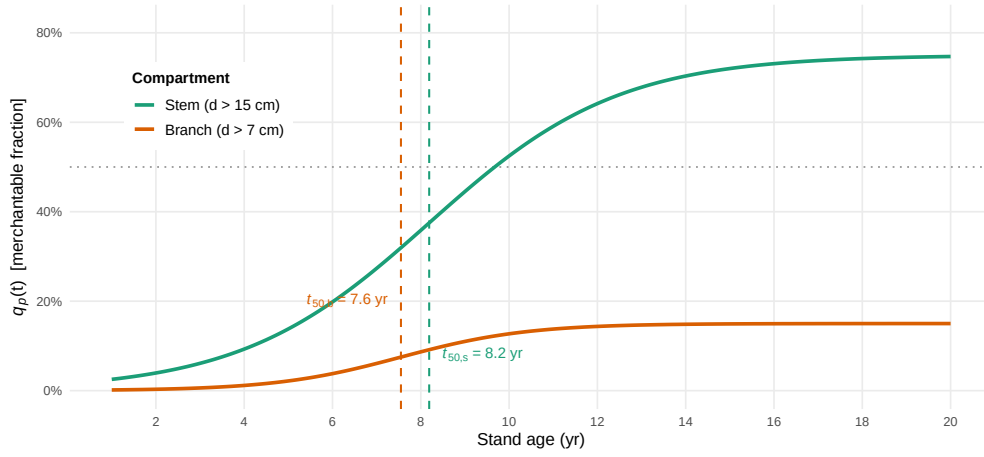


Figure 9: Logistic diameter-class merchantability functions $q_p(t)$ for stem ($d > 15$,cm, green) and branches ($d > 7$,cm, orange); the 50% threshold ages t_{50} are indicated by dashed vertical lines.

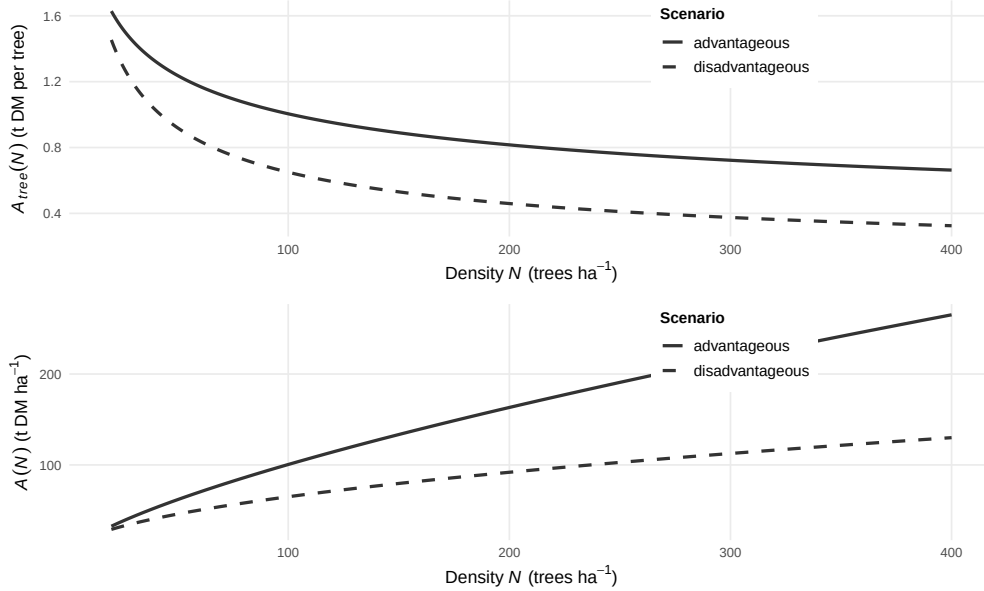


Figure 10: Tree-level and stand-level Gompertz asymptote functions $A_{tree}(N)$ and $A(N)$ for advantageous ($N = 4$, $\beta = -0.3$) and disadvantageous ($N = 6.5$, $\beta = -0.5$) site conditions.

Table 1: Notation for the biomass growth and harvest yield models.

Symbol	Description
$M(t)$	Total above- and belowground dry biomass per ha of AFS (t DM ha ⁻¹) at stand age t
$A(N)$	Stand-level Gompertz asymptote (t DM ha ⁻¹); density- and site-dependent
k	Gompertz growth-rate coefficient
t_0	Inflection age of stand biomass accumulation (years)
N	Stand density (trees ha ⁻¹)
C_{site}	Site-quality constant (kg ha ^{β-1}); conservative: 4 475, good site: 6 471
β	Competition-density exponent; fitted $\hat{\beta} = 0.380$
$f_p(t)$	asymptotic fraction of AGB attributed to compartment $p \in \{s, b, r\}$ at age t
$q_p(t)$	Merchantable fraction of compartment p biomass at age t (logistic)
r_p	Logistic growth rate for merchantability of compartment p
$t_{50,p}$	Age at which 50% of compartment p biomass is merchantable (yr)
$\eta_p(t)$	Merchantable dry biomass of compartment p per ha at harvest age t (t DM ha ⁻¹)
ω	Moisture content at felling (mass fraction); $\omega = 0.55$

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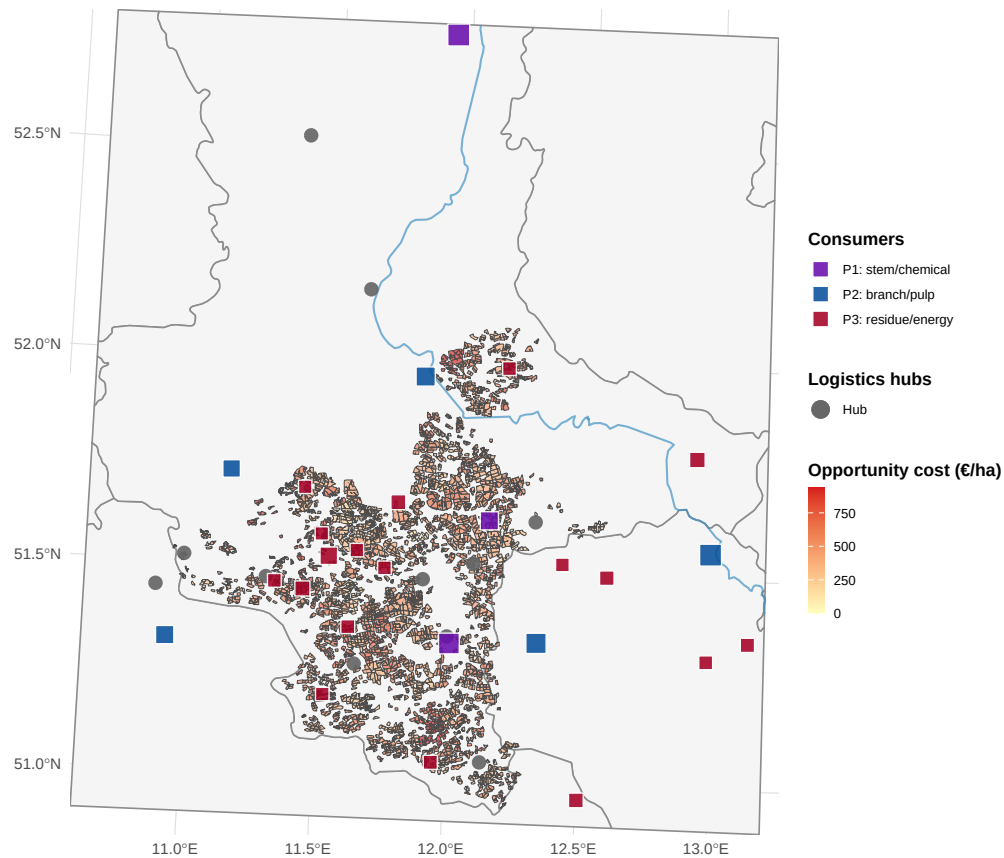


Figure 11: AFS supply chain network in Saxony-Anhalt. Grey lines: German federal state boundaries. Blue lines: major rivers. Coloured polygons: potential AFS parcels grouped by site (colours encode cluster membership; no legend). Orange symbols: pre-treatment facilities (circle = sawmill, triangle = logistics hub, size proportional to processing capacity). Coloured squares: consumer sites (violet: P1 chemical/pulp, blue: P2 pulp/paper, red: P3 energy/biogas, size proportional to total demand).

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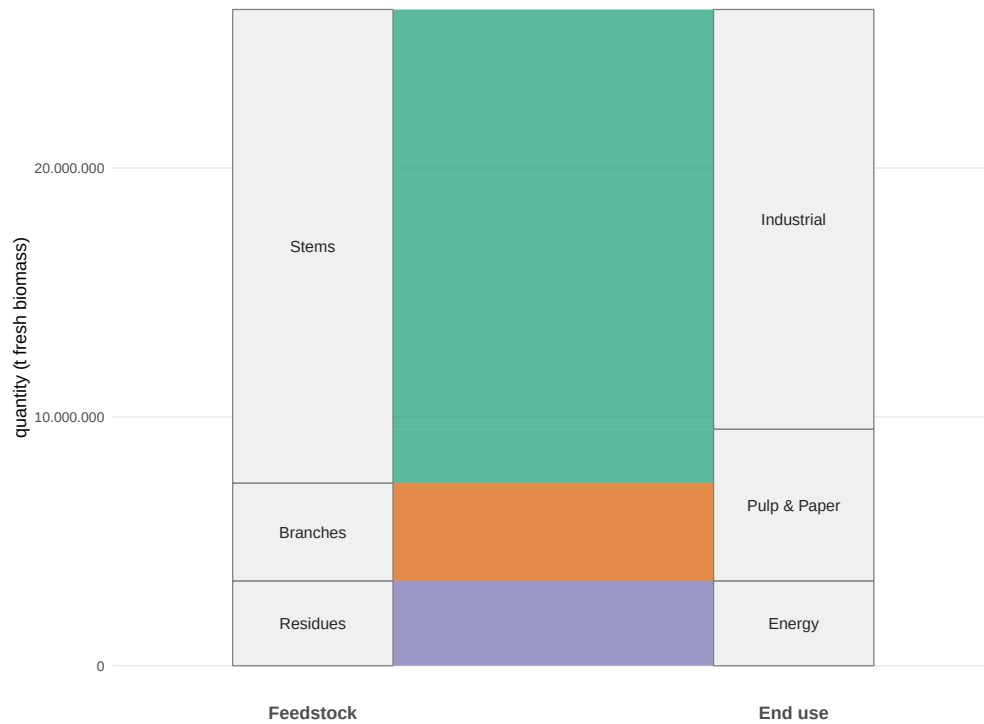


Figure 12: Total production quantity of biomass types and its industrial usage over planning horizon (in tons).

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Table 2: Notation for the MILP model.

Sets & Indices	
$i \in \mathcal{I}$	Agroforestry sites
$j \in \mathcal{J}$	Storage / pre-treatment facilities
$k \in \mathcal{K}$	Consumer sites
$p \in \mathcal{P}$	Product types (stem, branches, residues)
$(p, p') \in \mathcal{Q}$	Admissible product-substitution pairs (quality cascade)
$\mathcal{T} = \{1, \dots, T\}$	Planning periods (years)
$\mathcal{T}^{\text{harv}} = \{A^{\min} + 1, \dots, T\}$	Feasible harvest periods
$\mathcal{S}^{\text{est}} = \{(0, t) \mid t \in \{1, \dots, T - A^{\min}\}\}$	Establishment arcs: dummy source to period t
$\mathcal{S}^{\text{harv}} = \{(s, t) \mid s, t \in \mathcal{T}, t - s \in \{A^{\min}, \dots, A^{\max}\}\}$	Harvest cycles within age-lag bounds
$\mathcal{S}^{\text{term}} = \{(s, T + 1) \mid s \in \{A^{\min} + 1, \dots, T\}\}$	Termination arcs to dummy sink $T + 1$
$\mathcal{S} = \mathcal{S}^{\text{est}} \cup \mathcal{S}^{\text{harv}} \cup \mathcal{S}^{\text{term}}$	Age-lag arc set (establishment $\cup$ harvest $\cup$ termination)
Decision Variables	
$z_{ist} \geq 0$	Area of site i with AFS operation between s and t
$X_{ijpt} \geq 0$	Transport flow (t) of p from site i to facility j in t
$X_{jkpp't} \geq 0$	Flow of p from facility j to consumer k satisfying demand p' in t
$S_{jpt} \geq 0$	Inventory (t) of product p at facility j end of period t
Parameters	
T	Length of planning horizon (years)
$A^{\min}, A^{\max}$	Minimum / maximum stand age at harvest (years)
AREA_i	Area of site i (ha)
η_{pa}	Yield coefficient (t/ha) of fraction p at stand age a , derived from (4)
R_{kp}	Revenue (€/t) for product p at consumer k
C_i^{est}	Establishment cost (€/ha) at site i
C_i^{opp}	Annual opportunity cost (€/ha) at site i
C_i^{harv}	Harvest & refitting cost (€/ha) at site i
$c^{\text{tr-raw}}$	Transport cost rate (€/t km) for raw biomass
$c^{\text{tr-pre}}$	Transport cost rate (€/t km) for pre-treated biomass
c_j^{stor}	Storage holding cost (€/t) at facility j
d_{ij}, d_{jk}	Distance (km) between nodes
$\text{CAP}_j^{\text{stor}} / \text{CAP}_j^{\text{proc}}$	Storage and processing capacity (t) at facility j
D_{kpt}^{max}	Maximum demand (t) of fraction p at consumer k in period t

Table 3: Parameter values for the AFS-SCDcase study. Transport costs refer to road distance

Parameter	Description	Base-case value	Unit
Index sets			
$ \mathcal{T} $	Planning periods	40	yr
$ \mathcal{I} $	Sites	145	–
$ \mathcal{J} $	Pre-treatment / storage nodes	11	–
$ \mathcal{K} $	Consumers	28	–
$ \mathcal{P} $	Product grades	3	–
Biomass dynamics			
$A^{\min}$	Minimum harvest age (years)	3	yr
$A^{\max}$	Maximum stand age (years)	20	yr
Cost parameters			
c^{tr-raw}	Raw transport cost	0.08	€ t ⁻¹ km ⁻¹
c^{tr-pre}	Pre-processed transport cost	0.06	€ t ⁻¹ km ⁻¹
c_i^{est}	Establishment cost rate	2500	€ ha ⁻¹ yr ⁻¹
c_i^{harv}	harvest cost rate	150	€ ha ⁻¹ yr ⁻¹
c_i^{main}	Maintenance cost rate	10	€ ha ⁻¹ yr ⁻¹
c_i^{opp}	aver. opportunity cost	521	€ ha ⁻¹ yr ⁻¹
$\Delta, \delta_{pp'}$	conversion coefficients	1	–
Revenue parameters			
R_{P1}	Mean gate price P1	45	€ t ⁻¹
R_{P2}	Mean gate price P2	33	€ t ⁻¹
R_{P3}	Mean gate price P3	21	€ t ⁻¹